

Research Methodology in CS (CSEG3060_3)

Lecture 2: Criteria and Characteristics of a Good Research Problem

Unit 1: Introduction to Research Problem • Mapped Course Outcome: CO1

Lecture 2: Criteria and Characteristics of a Good Research Problem

Course: Research Methodology in CS (CSEG3060_3) **Program:** B.Tech (Computer Science)
Semester: 5 **Unit:** 1 — Introduction to Research Problem **Lecture:** 2 of Unit 1 **Mapped Course Outcome:** CO1 — *Understand the meaning, significance, and foundational components of research, including the formulation and evaluation of research problems.* **Estimated Reading Time:** 50–60 minutes

1. Lecture Overview

The formulation of a research problem is widely regarded as the most intellectually demanding phase of any empirical or analytical investigation. A well-formulated problem acts as a navigational compass for the entire research process: it dictates the choice of methodology, the nature of data to be collected, the analytical techniques to be applied, and the eventual contribution of the study to the body of knowledge. Conversely, a poorly formulated problem can lead to wasted effort, inconclusive findings, ethical complications, and a study that fails to gain acceptance in the academic community.

This lecture builds upon Lecture 1 (Understanding Research Problem: Meaning, Significance, Sources, and Identification) and transitions from the *conceptual* understanding of a research problem to its *evaluative* understanding. The focus is on establishing the benchmarks — both as external standards (criteria) and as intrinsic properties (characteristics) — that distinguish a scientifically rigorous and academically valuable research problem from one that is trivial, infeasible, or otherwise inadequate. Special emphasis is placed on the errors that researchers — particularly novice researchers — typically commit while selecting a research problem.

The lecture also extends the discussion into computer science by examining how these abstract criteria and characteristics manifest in concrete CS research scenarios such as algorithm design, machine learning, cybersecurity, software engineering, and data science.

2. Learning Objectives

Upon successful completion of this lecture, the student will be able to:

1. Define the concept of a "good" research problem and articulate why evaluation is necessary before commencing inquiry.
 2. Enumerate and explain the essential criteria for evaluating a research problem.
 3. Describe the distinguishing characteristics that a well-formulated research problem must inherently possess.
 4. Differentiate between *criteria* (external standards of evaluation) and *characteristics* (intrinsic properties) of a research problem.
 5. Identify the common errors committed by researchers during the selection of a research problem.
 6. Apply the criteria and characteristics to assess the quality of sample research problems drawn from computer science domains.
 7. Critically self-evaluate a proposed research problem using a structured checklist.
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3. Prerequisites

A clear understanding of the following concepts from Lecture 1 is assumed:

- The meaning of a research problem.
- The significance of the research problem in the research process.
- Sources from which research problems originate (literature, observation, experience, supervisors, conferences, etc.).
- The preliminary steps of identifying a research problem.

Familiarity with the structure of a research paper (IMRaD: Introduction, Methods, Results, and Discussion) and basic research terminology is also helpful.

4. Key Concepts and Definitions

The following definitions establish the conceptual foundation for the lecture:

- **Research Problem:** A specific intellectual or practical issue that a researcher seeks to investigate, analyse, or resolve through systematic scientific inquiry. It is the precise question that the research seeks to answer.

- **Criteria of a Research Problem:** The set of external standards, benchmarks, or yardsticks against which a research problem is evaluated or judged for its adequacy before approval and execution.
- **Characteristics of a Research Problem:** The inherent, internal qualities or properties that the research problem itself must possess in order to be considered scientifically valid.
- **Feasibility:** The practical possibility of conducting the proposed research within the constraints of time, resources, expertise, and access to data or subjects.
- **Novelty (Originality):** The extent to which the research problem contributes new knowledge, a new perspective, or a new solution to an unresolved or inadequately resolved question.
- **Researchability:** The degree to which the problem can be empirically or analytically investigated using available scientific methods, producing measurable or interpretable results.
- **Scope:** The clearly defined boundaries that delineate what the research will and will not cover.

Note on Terminology: While some textbooks treat "criteria" and "characteristics" as interchangeable, this lecture adopts the more rigorous distinction in which *criteria* are the evaluative lenses applied from outside, and *characteristics* are the inherent attributes possessed by the problem itself. This distinction is critical for examination purposes.

5. Detailed Exposition

5.1 The Meaning of a "Good" Research Problem

A research problem is considered "good" when it is simultaneously:

- Worth investigating (it addresses an important gap, conflict, or need).
- Investigable (it can be studied empirically or analytically using rigorous methods).
- Manageable (it can be completed within realistic resource, time, and expertise constraints).
- Generative (it has the potential to produce meaningful, publishable, or applicable results).

The goodness of a research problem is not an absolute property; it is relative to the researcher's training, institutional context, available resources, and the current state of the discipline. A problem that is excellent for a doctoral candidate may be unsuitable for an undergraduate project, and vice versa.

5.2 Criteria of a Good Research Problem

Criteria function as evaluation standards. They are applied to the problem by the researcher, supervisor, review committee, or funding body. The following criteria are widely accepted in the methodology literature:

5.2.1 Novelty and Originality

The problem must address a gap in existing knowledge. Pure replication studies, unless explicitly justified as replication research, do not constitute original research. Novelty may manifest as:

- Discovery of a new phenomenon.
- Development of a new method, algorithm, or tool.
- Application of an existing method to a new domain.
- Refutation or substantial modification of an existing theory.
- Deeper understanding of an existing phenomenon through novel data or analytical techniques.

In computer science, novelty is often demonstrated through a comparative performance evaluation against state-of-the-art baselines on benchmark datasets.

5.2.2 Feasibility

A good research problem must be practically possible to execute. Feasibility is assessed across multiple dimensions:

Dimension	Illustrative Question
Technical	Are the required tools, software, hardware, and datasets accessible?
Financial	Is the budget sufficient for computation, data acquisition, or participant recruitment?
Temporal	Can the research be completed within the stipulated timeline (e.g., one semester, three years)?
Cognitive	Does the researcher possess (or can acquire) the requisite statistical, programming, and domain knowledge?
Ethical	Will the research pass institutional ethics review?
Data	Are the required data sources accessible, or can they be generated within the project?

A problem that fails any one of these feasibility dimensions is unlikely to yield a successful study.

5.2.3 Relevance and Significance

The research problem must be relevant to the discipline and significant in its potential contribution. Significance is judged by:

- The size of the population or system affected.
- The severity of the problem being addressed.
- The generalizability of the expected findings.
- The potential to inform theory, policy, or practice.

In computer science, a problem on the scalability of distributed databases in cloud environments is highly relevant due to the widespread adoption of cloud computing across industries.

5.2.4 Clarity and Specificity

A good research problem must be stated clearly and unambiguously. The variables, the population/system under study, and the boundaries of the investigation must be evident. Vague statements such as "studying artificial intelligence" are not research problems; they are research domains. A clear specification, such as "evaluating the impact of attention mechanisms on the inference latency of transformer models in edge computing environments," meets the criterion of specificity.

5.2.5 Researchability (Investigability)

The problem must be amenable to systematic investigation. This requires that:

- The variables can be operationalised (translated into measurable indicators).
- The hypotheses (if any) can be tested.
- The data can be collected or generated.
- The methods of analysis are available and appropriate.

Some problems, though intellectually compelling, are not researchable. For instance, "What is the ultimate nature of consciousness?" is a philosophical problem that current scientific methods cannot resolve.

5.2.6 Ethical Acceptability

The research problem must not violate fundamental ethical principles: human dignity, privacy, informed consent, non-maleficence, and justice. In computer science, ethical considerations are particularly salient in research involving:

- Personal data of users (privacy).

- Surveillance systems.
- Algorithmic bias and fairness.
- Cybersecurity exploits.
- Dual-use AI capabilities.

5.2.7 Timeliness

The problem should be relevant to the current state of the discipline. A problem that was significant two decades ago but has since been thoroughly resolved is unlikely to be considered a good current research problem unless the researcher can articulate a clear new angle.

5.2.8 Manageability of Scope

The scope of the problem must be commensurate with the resources available. Overly ambitious problems that cannot realistically be completed violate the manageability criterion.

5.3 Characteristics of a Good Research Problem

Characteristics are the internal attributes the problem must possess. Once present, they make the problem amenable to evaluation by the criteria above.

5.3.1 It is Clearly Defined

The problem has well-articulated boundaries, specifies the population or system under study, and identifies the dependent and independent variables (where applicable). This characteristic is what allows the problem to be tested against the criterion of *clarity*.

5.3.2 It is Researchable

The problem, by its very nature, is one that can be subjected to scientific methods. It is not purely speculative, philosophical, or normative.

5.3.3 It is Significant

The problem, when resolved, has the potential to make a non-trivial contribution to theory, practice, or policy.

5.3.4 It is Novel

The problem is new, or it extends, refines, or re-examines existing knowledge from a fresh perspective.

5.3.5 It is Feasible

The problem is realistic given the researcher's competencies and the resources at hand.

5.3.6 It is Interesting and Motivating

A good research problem generates intrinsic curiosity in the researcher. Problems that the researcher finds boring or mechanical rarely produce passionate, high-quality work.

5.3.7 It Generates Testable Hypotheses or Answerable Questions

The problem leads naturally to specific questions or hypotheses that can be empirically or analytically addressed.

5.3.8 It is Consistent with the Researcher's Expertise

The problem aligns with the researcher's academic background, training, and prior work.

5.4 Distinguishing Criteria from Characteristics

The distinction is subtle but conceptually important.

Aspect	Criteria	Characteristics
Nature	External standards of judgment	Internal properties of the problem
Function	Used to evaluate the problem	Possessed by the problem
Examples	Feasibility, relevance, ethical acceptability	Clarity, novelty, researchability
Applied by	Reviewer, supervisor, funding body	Inherent to formulation

A useful analogy: a candidate appearing for an interview is judged against *criteria* (communication skills, technical knowledge), but the candidate must already *possess* the corresponding *characteristics* (articulation, knowledge). The interview evaluates; the attributes enable the evaluation.

5.5 Errors in the Selection of a Research Problem

Novice researchers frequently make the following errors. Recognising them is the first step toward avoiding them.

5.5.1 The Overly Broad Problem

The problem attempts to address too much within the scope of a single study. Example: "Developing a general-purpose artificial intelligence system." The scope is so vast that meaningful progress within a single project is impossible.

Correction: Narrow the problem to a specific sub-area, population, or application.

5.5.2 The Overly Narrow Problem

The problem is so restricted that its findings have negligible significance or generalizability. Example: "Investigating the typing speed of ten randomly selected students on a specific keyboard model on a specific day."

Correction: Broaden the problem to ensure that findings are generalisable or theoretically meaningful.

5.5.3 The Vague Problem

The problem is stated in such general terms that it is impossible to determine what is being investigated. Example: "Studying the impact of technology."

Correction: Operationalise the variables. Replace "technology" with the specific technology, population, and outcome variable of interest.

5.5.4 The Already Solved Problem

The problem has already been thoroughly addressed in the existing literature, and the proposed study does not offer a new angle.

Correction: Conduct a comprehensive literature review before finalising the problem. If the problem is indeed solved, identify a related unresolved sub-question.

5.5.5 The Trendy but Shallow Problem

The problem addresses a topic that is currently popular but lacks intellectual depth or genuine scientific contribution. Researchers sometimes select such problems to ride a wave of interest.

Correction: Substantiate the significance of the problem beyond its trendiness.

5.5.6 The Infeasible Problem

The problem requires resources, data, expertise, or access that the researcher does not have and cannot reasonably acquire.

Correction: Assess feasibility honestly and revise the problem to fit available resources.

5.5.7 The Ethically Problematic Problem

The problem, as formulated, raises serious ethical concerns that cannot be mitigated.

Correction: Redesign the problem to comply with ethical standards, or abandon it.

5.5.8 The Problem Without Theoretical Grounding

The problem is purely empirical and is not anchored in any theoretical framework. Such problems often lack a coherent rationale and produce findings that cannot be generalised or interpreted meaningfully.

Correction: Ground the problem in an established theoretical framework or develop a new conceptual model.

5.5.9 The Problem Dictated Entirely by the Supervisor

The researcher accepts a problem imposed by the supervisor without genuine intellectual engagement. This often leads to weak motivation and shallow contributions.

Correction: Negotiate ownership of the problem, ensuring alignment between the researcher's interests and the supervisor's expertise.

5.5.10 The Methodologically Inappropriate Problem

The researcher selects a problem that is interesting but cannot be adequately addressed by the chosen methodology (or by any available methodology).

Correction: Reconsider the problem, or identify an alternative methodology.

6. Criteria and Characteristics Applied to Computer Science Research Problems

To illustrate the practical application of the foregoing concepts, the following case studies examine three CS research problems — one exemplary, one problematic, and one borderline — using the criteria and characteristics discussed.

6.1 Case Study 1: A Good Research Problem

Title: *A Hybrid Deep Learning and Graph-Based Approach for Real-Time Detection of Zero-Day Vulnerabilities in Software-Defined Networks.*

Evaluation against criteria:

- *Novelty*: Yes — combines deep learning with graph-based analysis; addresses zero-day vulnerabilities in SDNs, both active areas.
- *Feasibility*: Yes — public SDN datasets (e.g., CICIDS, NSL-KDD) are available; deep learning frameworks are open-source; computation is manageable.
- *Relevance*: High — SDN security is a critical industry concern.
- *Clarity*: Yes — variables (network traffic features), population (SDN traffic), and outcomes (detection accuracy, false positive rate) are specified.
- *Researchability*: Yes — can be empirically evaluated against baseline models.
- *Ethical acceptability*: Yes — uses public datasets; no human subjects.
- *Timeliness*: Yes — SDNs and zero-day exploits are current.
- *Manageability*: Yes — appropriate for a Master's thesis or a multi-year doctoral investigation.

Evaluation against characteristics: The problem is clearly defined, researchable, significant, novel, feasible, and well-scoped. It is also likely to be motivating for a researcher with interests in cybersecurity and machine learning.

Verdict: This is a high-quality research problem.

6.2 Case Study 2: A Poorly Formulated Research Problem

Title: *Studying Computer Security.*

Evaluation against criteria:

- *Novelty*: Indeterminate — no specific gap identified.
- *Feasibility*: Indeterminate — scope is undefined.
- *Relevance*: Yes, in principle, but too broad.
- *Clarity*: Fails — the problem statement is too vague.
- *Researchability*: Fails — without specification, the problem cannot be operationalised.
- *Manageability*: Fails — the scope is enormous.

Diagnosis: This is a research *domain*, not a research *problem*. It exhibits every major error: vagueness, excessive breadth, and absence of operationalisation.

Verdict: This is not a research problem. It is the title of a textbook, not a thesis.

6.3 Case Study 3: A Borderline Problem

Title: *A Comparative Study of Sorting Algorithms.*

Evaluation against criteria:

- *Novelty:* Limited — sorting algorithms have been exhaustively studied. The problem must identify a new angle (e.g., performance under novel hardware constraints such as quantum processors or neuromorphic chips).
- *Feasibility:* Yes.
- *Clarity:* Yes — but potentially trivial if limited to standard algorithms on standard hardware.
- *Timeliness:* Only if the comparative angle addresses modern contexts.

Diagnosis: A problem that could be excellent if re-formulated with a novel angle (e.g., *Comparative Performance of Sorting Algorithms on Quantum Annealers*) but is poor if left as stated.

Verdict: A weak problem as stated; potentially strong if refined.

7. A Diagnostic Checklist for Evaluating a Research Problem

The following checklist may be used by students to self-evaluate a proposed research problem before submitting it for supervisor approval or institutional review:

1. Have I stated the problem in a single, precise sentence?
2. Have I clearly identified the population, system, or phenomenon under study?
3. Have I specified the variables or research questions?
4. Does the problem address a gap identified through a literature review?
5. Can the problem be empirically or analytically investigated within my timeframe?
6. Do I have (or can I acquire) the necessary data, tools, and expertise?
7. Has the problem passed ethical review?
8. Is the scope manageable?
9. Is the problem interesting enough to sustain my motivation?
10. Is the problem grounded in an appropriate theoretical or conceptual framework?

If any answer is "no" or "uncertain," the problem requires further refinement before approval.

8. Summary

This lecture has established the evaluative foundations of research problem formulation in computer science research methodology. The principal takeaways are:

- A "good" research problem is one that is *worth investigating*, *investigation-ready*, *manageable*, and *generative* of meaningful contribution.
- **Criteria** are the external benchmarks (novelty, feasibility, relevance, clarity, researchability, ethical acceptability, timeliness, manageability) used to judge the problem.
- **Characteristics** are the internal properties (clarity, researchability, significance, novelty, feasibility, interest, testability, expertise alignment) the problem itself must possess.
- The two concepts are interrelated: characteristics enable the problem to satisfy the criteria.
- The most common errors in selection are excessive breadth, excessive narrowness, vagueness, premature saturation of the literature, ethical infeasibility, and absence of theoretical grounding.
- In computer science, these evaluative standards are applied to problems spanning algorithms, machine learning, cybersecurity, software engineering, networks, databases, and emerging areas such as quantum computing and ethical AI.

A disciplined application of these criteria and characteristics substantially improves the probability that a research project will be completed successfully, contribute meaningfully to the discipline, and withstand peer review.

9. Exam Preparation Tips

1. **Master the distinctions.** Examiners frequently ask students to differentiate between criteria and characteristics. Memorise the table in Section 5.4 and reproduce it clearly in the answer.
2. **Use CS-specific examples.** Generic answers (drawn from social science examples) attract less credit. Always anchor your explanations in CS contexts (algorithms, ML, networking, security, databases, software engineering).
3. **Apply, do not just list.** When asked to evaluate a research problem, do not merely list the criteria. Demonstrate the application: *"This problem satisfies the criterion of feasibility because the NSL-KDD dataset is publicly available..."*
4. **Memorise the common errors.** Questions on errors in the selection of a research problem are very frequent. Know all ten errors listed in Section 5.5, and be able to give one CS-specific example for each.
5. **Practice problem diagnosis.** Take sample research titles and evaluate them against the checklist in Section 7. This is the most common style of exam question.

6. **Use bullet points and tables judiciously.** Examiners reward clarity of presentation. A well-structured table comparing criteria and characteristics can earn full marks even in a short-answer context.
 7. **Link to CO1.** The Course Outcome CO1 is satisfied by demonstrating an evaluative understanding of what constitutes a research problem. Phrase your answers to show that you understand not just *what* the criteria are, but *why* they matter for the integrity of the research process.
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10. Practice Questions

Short Answer Questions (2–3 marks each)

1. Define the term "criterion" in the context of evaluating a research problem. Give two examples.
2. Differentiate between a "criterion" and a "characteristic" of a research problem.
3. State any four common errors committed by researchers during the selection of a research problem.
4. Why is novelty considered an essential criterion of a good research problem?
5. What is meant by "manageability of scope" in research problem formulation?

Long Answer / Essay Questions (8–10 marks each)

1. Critically examine the criteria of a good research problem. Illustrate your answer with two examples from computer science research.
2. Discuss the characteristics that a research problem must inherently possess. How do these characteristics enable the problem to satisfy the evaluation criteria?
3. Identify and explain any six common errors in the selection of a research problem. For each error, provide a CS-specific example and suggest a corrective measure.
4. *"A good research problem is one that is both intrinsically valuable and methodologically tractable."* Discuss this statement in the context of computer science research.
5. Consider the following research problem: *"Improving the efficiency of search engines using artificial intelligence."* Evaluate this problem against the criteria discussed in the lecture. Suggest a refined formulation that addresses the identified weaknesses.

Applied / Diagnostic Questions

1. Below are three research titles. Identify which one is a well-formulated research problem and justify your choice using the criteria and characteristics discussed in the lecture.
 - (a) "A Study of Cloud Computing."

- (b) "Evaluating the Impact of Containerisation on the Cold-Start Latency of Serverless Functions in Multi-Cloud Environments."
 - (c) "Everything About Computers."
2. A PhD scholar wishes to study the fairness of facial recognition systems across demographic groups. Using the checklist in Section 7, evaluate whether this is a well-formulated research problem. Identify any gaps in formulation and propose refinements.
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11. Suggested Further Reading

1. Kothari, C. R., & Garg, G. (2019). *Research Methodology: Methods and Techniques* (4th ed.). New Age International Publishers. — Chapters 2 and 3 cover problem formulation and criteria of a good research problem in considerable depth.
 2. Creswell, J. W., & Creswell, J. D. (2018). *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches* (5th ed.). SAGE Publications. — Chapter 4 provides a contemporary treatment of problem statement formulation.
 3. Sekaran, U., & Bougie, R. (2016). *Research Methods for Business: A Skill-Building Approach* (7th ed.). Wiley. — Chapter 4 offers a clear exposition of criteria and characteristics.
 4. Bryman, A. (2016). *Social Research Methods* (5th ed.). Oxford University Press. — Provides a rigorous treatment of research problem formulation in the social and behavioural sciences, with transferable insights for CS.
 5. Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). *Experimentation in Software Engineering: An Introduction*. Kluwer Academic Publishers. — For CS-specific guidance on formulating research problems in software engineering.
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End of Lecture Notes — Lecture 2, Unit 1. Mapped Course Outcome: CO1. Prepared for academic instructional use, B.Tech (CS), Semester 5, UPES University.